Dajare Types and Individualised Sense of Humour - A Prelude to PUNDA Project -

Pawel Dybala　Rafal Rzepka　Kenji Araki
Graduate School of Information Science and Technology, Hokkaido University

Abstract:
In this paper, we present an outline of PUNDA Project, aimed to create a Japanese pun generating system. We describe first two steps of the Project – extraction of *Dajare* Types Set and an idea of individualised sense of humour.

1. Introduction

Pun generating systems become subjects of some research project of AI or natural language processing study. However, machine sense of humour in general is still rather neglected field of study. In this paper, we propose an improved version of Japanese pun's types and a possible method of individualising a model of humour sense. These two issues are the preliminary steps of PUNDA Project, designed to create a functional Japanese pun generating engine.

2. What is PUNDA Project?

PUNDA (PUN-*DAjare*) Project's main goal is constructing an engine that would be able to recognise and generate Japanese puns called *dajare*. The project includes the following steps:
a) ***Dajare* Types Set extraction**
b) **Preliminary research** (method of individualising model of linguistic humour sense)
c) **Main research** (constructing a pun generating module)
d) **Improvement and evaluation**
e) **Equipping a non-task oriented conversational system with a pun generating module**
In this paper, we describe the first two steps, which will provide a solid for the main part of our research.

3. *Dajare* Types Set

In previous researches on pun generating systems, few mechanisms of linguistic jokes were extracted. Some of them focused only on punning riddles [1], while other presented a wider, but still insufficient range of schemes [2]. In our PUNDA Project we are going to use *Dajare* Types Set, based on *dajare* classification proposed in our earlier thesis [3]. We distinguished twelve *dajare* phonotactic types, basing on moras as the most appropriate metric units for Japanese language [4]. These groups are:

a) **Homophony**
(カエルが帰る *Kaeru ga kaeru* <The frog comes back>)
b) **Mora addition**
- Initial mora
(スイカは安いか *Suika wa yasuika* <Watermelon is cheap>)
- Final mora
(カバのかばん *Kaba no kaban* <Sea cow's bag>)
- Internal mora
(布団が吹っ飛んだ *Futon ga futtonda* <Futon flew away>)
c) **Mora omission**
- Final mora
(スキーが好き *Sukii ga suki* <I like skiing>)
- Internal mora
(ステーキはすてき *Suteeki wa suteki* <Steaks are cool>
d) **Mora transformation**
- Consonant transformation
(目だまし時計 *Medamasidokei* <Eye-misleading clock>)
- Vowel transformation
(背中の悲劇 *Senaka no higeki* <Senaka's tragedy>, senaka = back[1])
e) **Phoneme and syllable metathesis**
(漫画を読むのはま、我慢 *Manga wo yomu no wa ma, gaman* <I can stand reading commics>)

[1] Seneka – a Roman philosopher and dramatist (ca. 4 BC–AD 65)

f) *Kanji* reading change
(食王 *Syokkingu* <Shocking>)
g) Morpheme metathesis
(男を売る思い出 *Otoko wo uru omoide* <A memory of selling a man>[2])
h) Blend
(老いてはことをし損ずる *Oite wa koo wo sisonzuru* <When you get old, you start to make waste>[3])
i) Division
(ゆで卵をゆでたのは孫 *Yudetamago wo yudeta no wa mago* <It's the grandchild who boiled the egg>)
j) Quiz
(タヌキの金玉はなんと言いますか？ – 「キンマ」 *Tanuki no kintama wa nan to iimasuka? – Kinma* <How do you call tanuki's balls? – Betel>)
k) Mix of Japanese and foreign languages
(戸部君、ハムレットが君に言ってるだろう、トベ・オル・ノットベ *Tobe-kun, Hamuretto ga kimi ni itte iru darou, tobe oru nottobe* <Tobe, Hamlet is talking to you – To be or not to be>)
l) Pause transference
(金をくれ、頼む！・金をくれた、飲む！ *Kane wo kure, tanomu! –Kane wo kureta, nomu!* <Please, give me some money! –You gave me the money, let's drink!>)

This classification has been based on the set of collected human generated *dajare*. Some of these groups also cover the changes in meaning, which will be used in further steps of the Project [3].

4. Preliminary research – towards individualised sense of humour

An important attribute of the phenomenon of humour sense is its individuality. In fact, every humour - equipped human being develops a unique sense of humour, which allows him (or her) to decide whether something is funny or not (although there are many theories of humour in general, most of them seem too general to be used in a pun generating system [5]). There are many factors that influence our perception of funniness. Among them, age, sex, education level, interests or social position seem to be of significant importance.

We propose to conduct a survey to check what kind of *dajare* are considered to be funny in general, and what is the relation between funniness and factors mentioned above. The questionnaire will include jokes from all twelve categories from the *Dajare* Types Set, and will be sent to representatives of different social groups. This will allow us to create several models of linguistic humour sense, which will lead to the individualisation of the system.

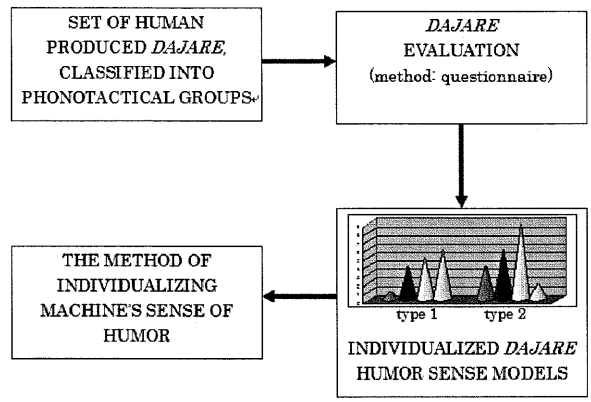


Fig.1. Sense of humour individualisation

5. Summary

Creating the *Dajare* Types Set and evaluation of human-generated puns both play an important role in PUNDA Project. Their completion will provide us with a solid basis to conduct the main part of research and, in consequence, allow creating a *dajare* generating engine, which would be more functional than existing systems.

Reference

[1] Kim Binsted, Osamu Takizawa, "Computer generation of puns in Japanese", Sony Computer Science Lab, Communications Research Laboratory, September 1997
[2] Koichiro Tanabe, "Pun Processing in a computer - A Primary Design for a Pun Generator", Sanno University Bulletin Vol.26 No. 1 September 2005
[3] Pawel Dybala, "*Dajare* – Nihongo ni okeru dōonigi ni motozuku gengo yūgī", Jagiellonian University, 2006 (http://tnij.org/mastert)
[4] Haruo Kubozono, „Nihongo no onsei", Iwanami Shoten, 1999
[5] Kim Binsted, "Machine humour: An implemented model of puns", University of Edinburgh, 1996
[6] Yūshi Odashima, "*Dajare* no ryūgī", Kodansha, 2000

[2] From the title of TV drama「思い出を売る男」*Omoide wo uru otoko* <A man selling memories>
[3] A blend of two proverbs:「急いてはことを仕損ずる」*Seite wa koto wo sisonzuru* <Haste makes waste> and「老いては子に従え」*Oite wa ko ni sitagae* <When you get old, you should listen to your children>